

7.3 Halogen Compounds (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	7. Organic Chemistry (A Level Only)
Topic	7.3 Halogen Compounds (A Level Only)
Difficulty	Hard

Time allowed: 30

Score: /22

Percentage: /100

Question 1a

Nitration of a benzene ring forms nitrobenzene. The subsequent reactions of nitrobenzene are shown in Fig. 1.1.

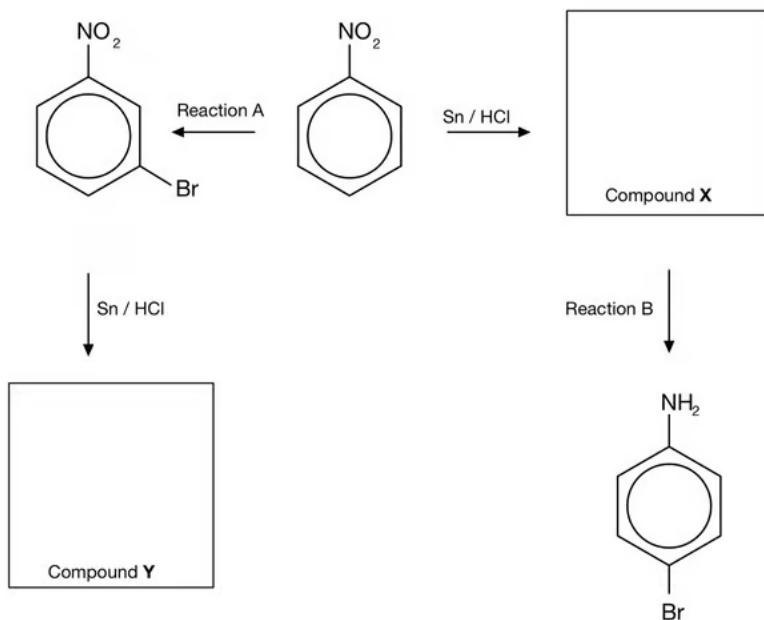


Fig. 1.1

The structures of the products have the same molecular formula. Explain why these products are different. In your answer include the structures of compounds **X** and **Y** and reagents for reactions **A** and **B**.

[5 marks]

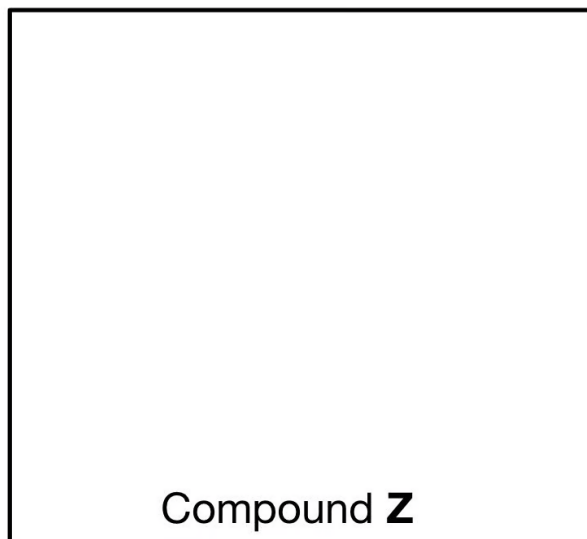
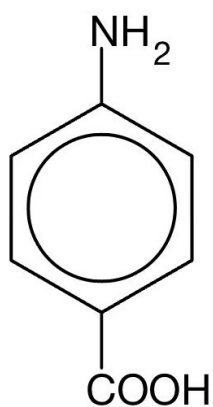
Question 1b

Using curly arrows, describe the mechanism for reaction **A**.

[3 marks]

Question 1c

State the IUPAC name of Compound **Z** and draw the structure.



[2 marks]

Question 2a

Chlorobenzene is formed by bubbling chlorine gas into benzene at room temperature in the presence of anhydrous AlCl_3 .

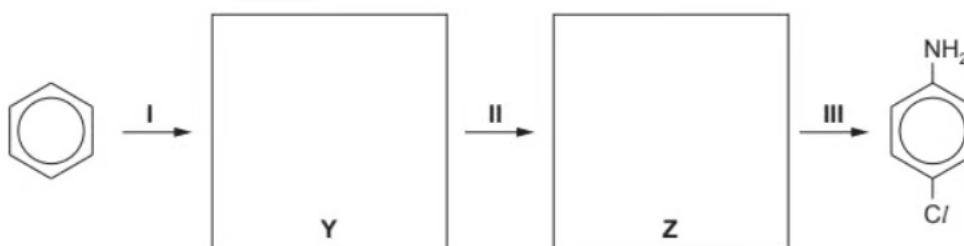
Explain, by the use of equations, how AlCl_3 acts as a catalyst in the chlorination of benzene.

[4 marks]

Question 2b

Nitrobenzene undergoes further substitution considerably more slowly than chlorobenzene. In nitrobenzene, the incoming group joins the benzene ring in the 3-position, whereas in chlorobenzene the incoming group joins to the benzene ring in the 4-position.

Use these ideas to suggest the structures of the intermediate compounds **Y** and **Z** in the following synthesis of 4-chlorophenylamine.



[1 mark]

Question 2c

Suggest the structure of the organic product formed when 4-chlorophenylamine reacts with bromine water.

[1 mark]

Question 3a

Fig. 3.1 shows two reactions of methyl benzene.

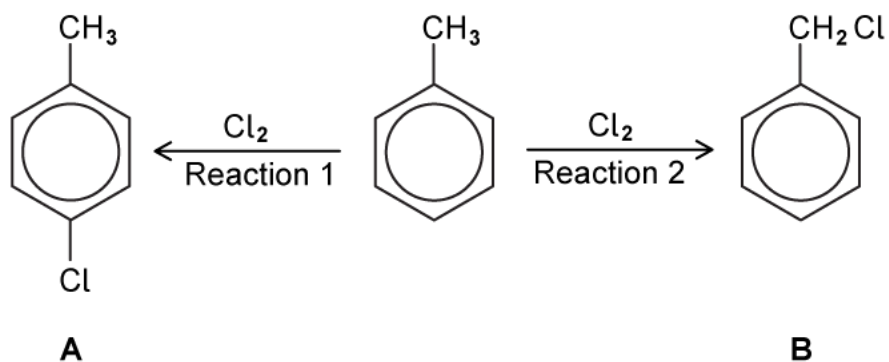


Fig. 3.1

- i)
State the conditions required for reaction 1.

[1]

- ii)
State the conditions required for reaction 2.

[1]

[2 marks]

Question 3b

State the reagent needed to carry out the following reaction occurring in Fig. 3.2.

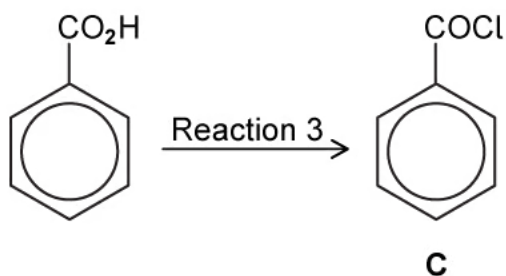


Fig. 3.2

reagent for reaction 3.....

[1 mark]

Question 3c

The three chloro-compounds **A**, **B** and **C** vary in their ease of hydrolysis.

i)

Place a tick in the box in Table 3.1 corresponding to the correct relative rates of hydrolysis. [the symbol '>' means 'faster than'].

Table 3.1

	Place one tick only in this column
A > B > C	
A > C > B	
B > A > C	
B > C > A	
C > B > A	
C > A > B	

[1]

ii)

Suggest an explanation for these differences in reactivity

[2]

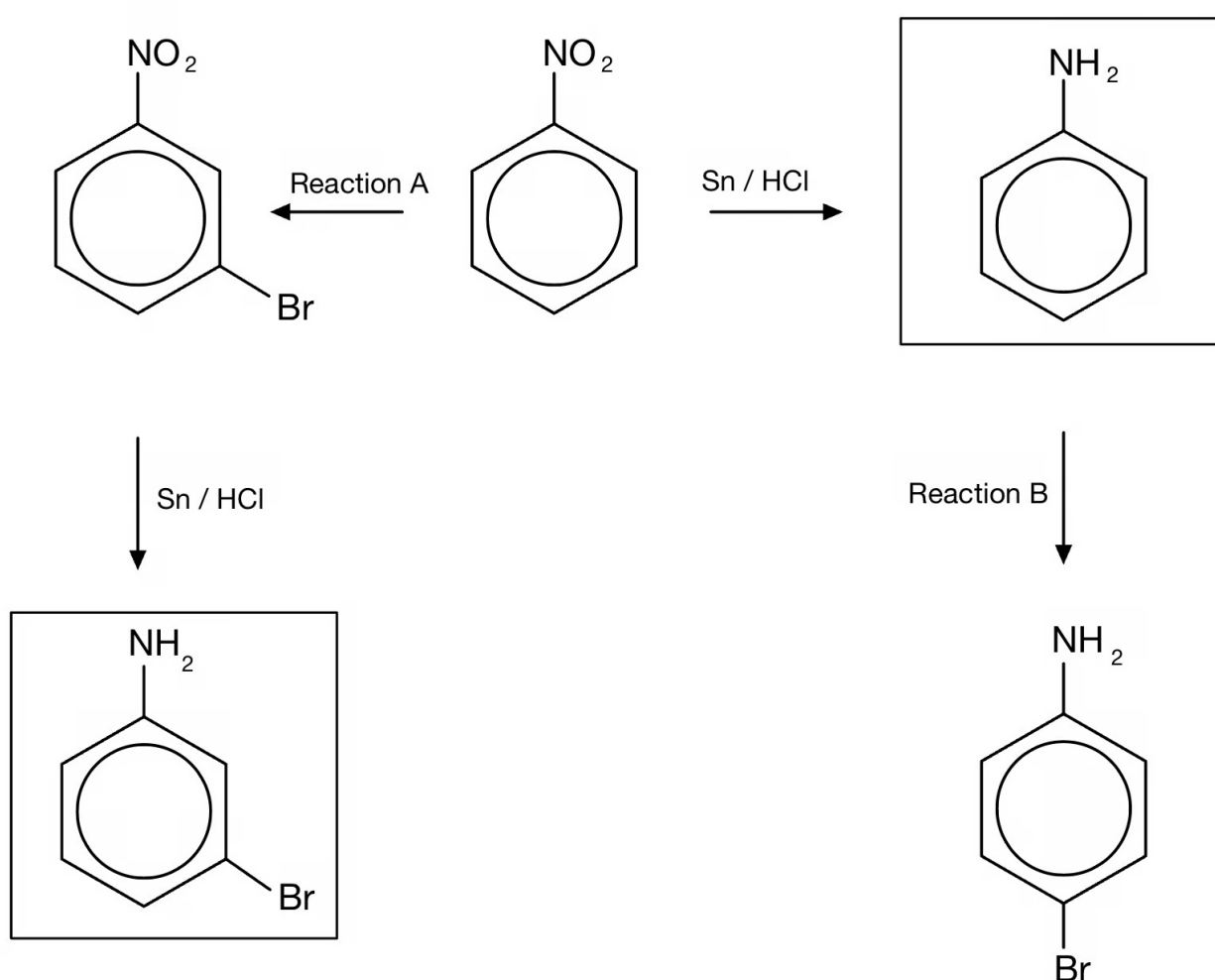
[3 marks]

Model Answers: Hard

1a

a) The structures, reagents and reason why the two different compounds are different are:

Structures:



- Correct structure of compound X; [1 mark]
- Correct structure of compound Y; [1 mark]

Reagents:

- Reaction **A** and **B** reagents for bromination = bromine / Br₂ in presence of (anhydrous) aluminium bromide / AlBr₃; [1 mark]

Reasons:

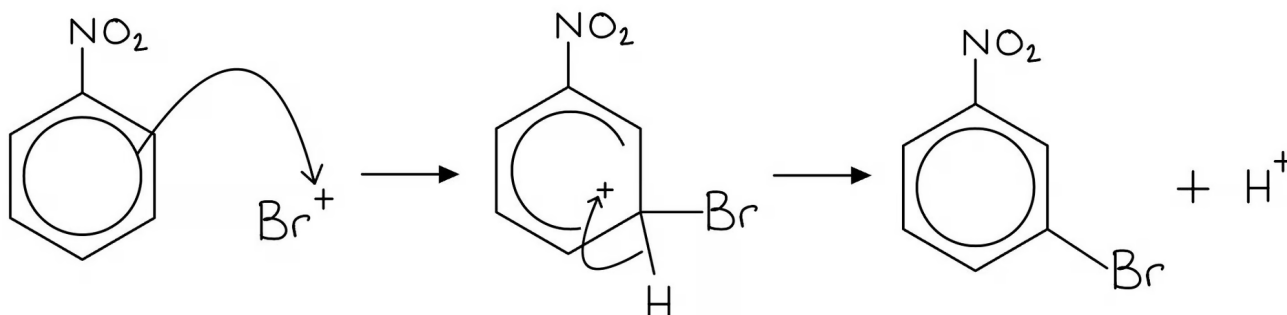
- Reaction **A** should take place before reduction as the NO₂ group is 3 directing; [1 mark]
- Reduction should take place before reaction **B** as the NH₂ group is 2 / 4 directing; [1 mark]

[Total: 5 marks]

- This question tests your knowledge of directing effects on the benzene ring
- It may appear that the reaction is the same, but in fact, they must take place in a certain order due to the directing effects of the different groups
 - NO₂ directs to the 3- position
 - NH₂ directs to the 2- and 4- positions
- We are told the products are different but have the same molecular formula, therefore the substitutions on the benzene ring will have occurred at different places
- As a general rule: (you do not need to be able to explain why)
 - Electron-donating groups make the ring more reactive, and direct into the 2- and 4- positions
 - Electron-withdrawing groups make the ring less reactive and direct into the 3-position

1b

b) The mechanism for Reaction **A** is:



- Curly arrow from the ring inside benzene to the Br^+ electrophile; [1 mark]
- Correct intermediate structure; [1 mark]
- Curly arrow from the C-H bond back to the + within the ring

AND

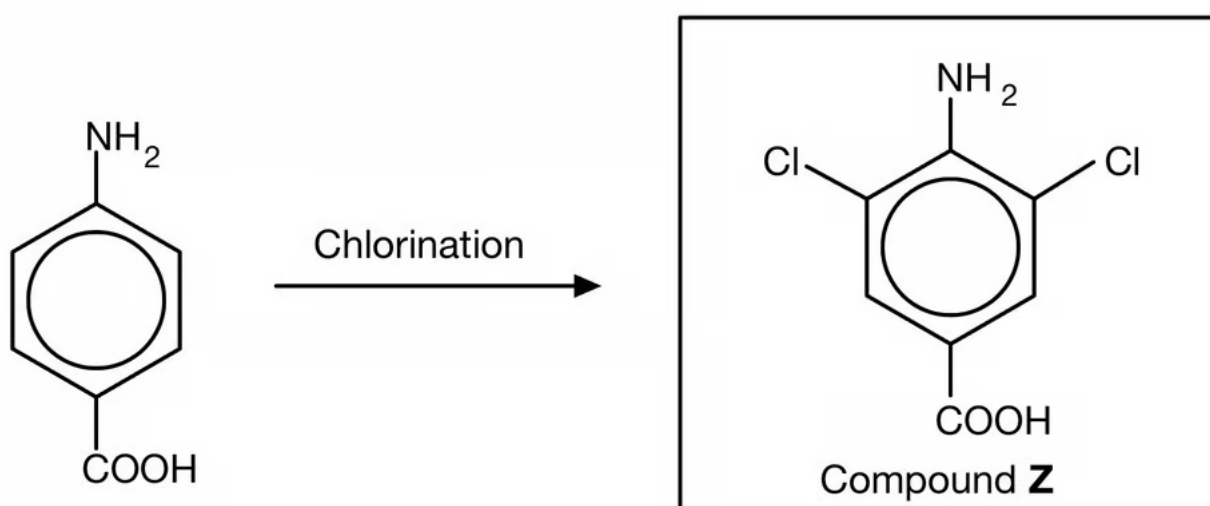
Correct products shown; [1 mark]

[Total: 3 marks]

- The bromine is being substituted onto the 3rd carbon, therefore the 'horseshoe' must be located around this carbon
- The horseshoe must cover at least half of the ring with the 'gap' at the correct C

1c

c) The name and structure of compound **Z** are:



- 4-amino-3,5-dichlorobenzoic acid; [1 mark]
- Correct structure of compound **Z**; [1 mark]

[Total: 2 marks]

- NH_2 is an electron-donating substituent and will direct the incoming electrophile to attack the 2- and 4- positions (in relation to the $-\text{NH}_2$ group)
 - The 4- position is already substituted with $-\text{COOH}$, so no substitution occurs in this position
- $-\text{COOH}$ is an electron-withdrawing substituent and will direct the incoming

electrophile to attack the 3- and 5- positions (in relation to the -COOH group)

- The name is based on benzoic acid, so the position of the -COOH group is assigned the number 1 position

2a

a) AlCl_3 acts as a catalyst in the chlorination of benzene by:

- $\text{Cl}_2 + \text{AlCl}_3 \rightarrow \text{Cl}^+ + \text{AlCl}_4^-$; [1 mark]
- $\text{Cl}^+ + \text{C}_6\text{H}_6 \rightarrow \text{C}_6\text{H}_5\text{Cl} + \text{H}^+$; [1 mark]
- $\text{H}^+ + \text{AlCl}_4^- \rightarrow \text{HCl} + \text{AlCl}_3$; [1 mark]
- AlCl_3 is used in the first step

AND

AlCl_3 is regenerated in the final step; [1 mark]

[Total: 4 marks]

- You need to give an equation to show how AlCl_3 is used to produce the electrophile, Cl^+
- The second step shows the electrophilic substitution of benzene to form chlorobenzene
- In the final step, you need to show how AlCl_3 is regenerated
- The overall equation for the reaction is:
 - $\text{Cl}_2 + \text{C}_6\text{H}_6 \rightarrow \text{C}_6\text{H}_5\text{Cl} + \text{HCl}$

2b

b) The structures of **Y** and **Z** are:



...the number 1 position, as the position the -OH group is assigned the number 1 position

2a

a) AlCl_3 acts as a catalyst in the chlorination of benzene by:

- $\text{Cl}_2 + \text{AlCl}_3 \rightarrow \text{Cl}^+ + \text{AlCl}_4^-$; [1 mark]
- $\text{Cl}^+ + \text{C}_6\text{H}_6 \rightarrow \text{C}_6\text{H}_5\text{Cl} + \text{H}^+$; [1 mark]
- $\text{H}^+ + \text{AlCl}_4^- \rightarrow \text{HCl} + \text{AlCl}_3$; [1 mark]
- AlCl_3 is used in the first step

AND

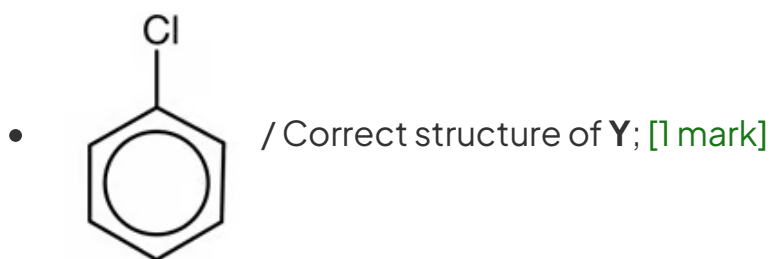
AlCl_3 is regenerated in the final step; [1 mark]

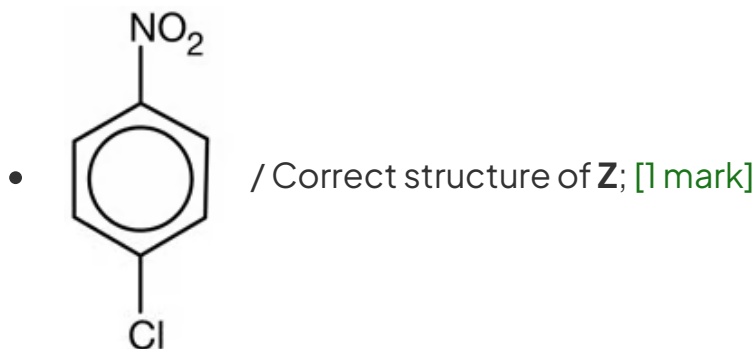
[Total: 4 marks]

- You need to give an equation to show how AlCl_3 is used to produce the electrophile, Cl^+
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- The overall equation for the reaction is:
 - $\text{Cl}_2 + \text{C}_6\text{H}_6 \rightarrow \text{C}_6\text{H}_5\text{Cl} + \text{HCl}$

2b

b) The structures of **Y** and **Z** are:



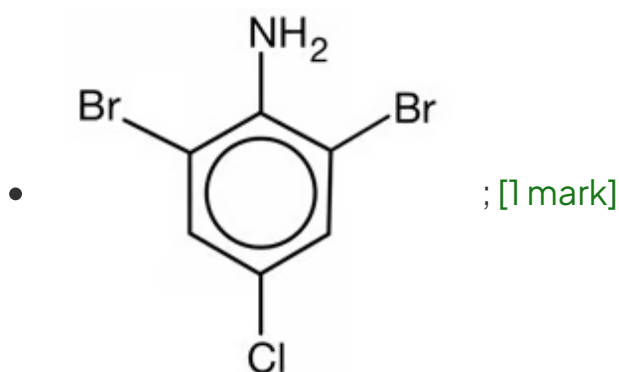


[Total: 2 marks]

- **Y** is chlorobenzene and **Z** is 4-chloronitrobenzene;
- To produce the final product, the following needs to occur:
 - Chlorine needs to be substituted onto the ring
 - $-\text{NO}_2$ needs to be substituted and then reduced to $-\text{NH}_2$
- To decide which is substituted first, Cl or NO_2 , consider the positions of Cl and NH_2 in the final product
- Cl must have been substituted first so that NO_2 is then directed into the 4-position
 - If NO_2 had been substituted first, Cl would have been directed into the 3-position, which would result in 3-chloronitrobenzene being formed as the final product

2c

c) The structure of the organic product formed when 4-chlorophenylamine reacts with bromine water is:



[Total: 1 mark]

- You would be awarded the mark for a structure that had 2 bromine atoms

substituted in any combination of the available positions

3a

a)

i) The conditions required for reaction **1** are:

- $\text{AlCl}_3 / \text{FeCl}_3 / \text{Al/Fe/I}_2$ (+ heat); [1 mark]

ii) The conditions required for reaction **2** are:

- (Sun)light / hf / UV; [1 mark]

[Total: 2 marks]

- Reaction **1** substitutes a hydrogen atom on the benzene ring for a chlorine atom
 - This is an example of electrophilic substitution which requires a halogen carrier
 - **Careful:** You must use chlorine in the halogen carrier and not another halogen such as AlBr_3
- Reaction **2** substitutes a hydrogen atom on the methyl group, not the benzene group
 - This is an example of free radical substitution, a reaction that requires ultra-violet light
 - **Careful:** Stating aqueous in your answer will not gain the mark as this is not a correct condition

3b

b) The correct reagent is:

- SOCl_2
OR
 PCl_3
OR
 PCl_5 ; [1 mark]

[Total: 1 mark]

- This reaction involves the formation of an acyl chloride from a carboxylic acid

- This reaction can be completed by the addition of:
 - **Solid** phosphorus(V) chloride (PCl_5)
 - **Liquid** phosphorus(III) chloride (PCl_3) and heat
 - **Liquid** sulfur dichloride oxide (SOCl_2)

3c

i) The correct order for ease of hydrolysis is:

- **C > B > A**; [1 mark]

ii) The reasons for the difference in hydrolysis are:

- (Acyl chloride fastest) highly δ^+ carbon atom joined to 2 electronegative atoms
OR
Addition-elimination mechanism is possible; [1 mark]
- (Aryl chloride slowest) delocalisation of lone pair over ring stronger C-Cl bond; [1 mark]

[Total: 3 marks]

- Hydrolysis involves the nucleophile OH^- attacking the compound
- Halogenoarenes, such as compound **A**, do not readily undergo nucleophilic substitution reactions unless there are harsh conditions
- The unreactivity of halogenoarenes can, therefore, be explained by the delocalisation of a lone pair on the halogen over the benzene
- This causes additional stabilisation of the system and strengthens the carbon-halogen bond, which affects the reactions that halogenoarenes will undergo
- Even though compounds **B** and **C** contain a benzene ring, the OH^- nucleophile can attack the δ^+ carbon atom relatively easily
- **C** is more reactive as the C-Cl bond is more polar than the C=O bond in compound **B** due to the higher difference in electronegativity